

Sulforaphane mobilizes cellular defenses that protect skin against damage by UV radiation

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UV radiation (UVR) is a complete carcinogen that elicits a constellation of pathological events, including direct DNA damage, generation of reactive oxidants that peroxidize lipids and damage other cellular components, initiation of inflammation, and suppression of the immune response. Recent dramatic increases in the incidence of nonmelanoma skin cancers are largely attributable to higher exposure of an aging population to UVR. Therefore, the development of cellular strategies for intrinsic protection of the skin against the deleterious effects of UVR is imperative. Here we show that erythema resulting from UVR is a comprehensive and noninvasive biomarker for assessing UVR damage and can be precisely and easily quantified in human skin. Topical application of sulforaphane-rich extracts of 3-day-old broccoli sprouts up-regulated phase 2 enzymes in the mouse and human skin, protected against UVR-induced inflammation and edema in mice, and reduced susceptibility to erythema arising from narrow-band 311-nm UVR in humans. In six human subjects (three males and three females, 28–53 years of age), the mean reduction in erythema across six doses of UVR (300–800 mJ/cm² in 100 mJ/cm² increments) was 37.7% (range 8.37–78.1%; $P = 0.025$). This protection against a carcinogen in humans is catalytic and long lasting.